

Gravitas: A Browser-Based Astrophysics Laboratory for Prediction, Measurement, and Discovery

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ABSTRACT

Astronomy students are often asked to reason about systems that cannot be handled in a laboratory, evolve over inaccessible timescales, or reveal themselves only through indirect measurements. GRAVITAS is a free, browser-based astrophysics environment designed to turn those limitations into opportunities for experimentation. Its planned version 1.0.0 release combines a configurable gravitational sandbox with 59 ready-to-run scenarios and 22 guided investigations. Students can build and alter systems, change reference frames, measure distances, angles, periods, energies, forces, and orbital elements, and observe simulated systems through transit, radial-velocity, astrometric, rotation-curve, stellar-evolution, and gravitational-wave displays. Guided activities use a predict–test–measure–revise–explain cycle, retain the student’s evidence, and export a report for submission. Instructor guides, answer keys, short classroom routes, lecture and embed modes, reproducible share links, Spanish localization, keyboard support, and offline operation lower the practical barrier to adopting the simulation in introductory astronomy and physics courses. This article describes the learning experiences available in GRAVITAS, illustrates several ways it can be used from a brief demonstration to a full laboratory exercise, and states the scientific scope and limitations of the models. The aim is not to replace quantitative calculation or observation, but to give students a place where they can make a prediction, change one thing, measure what follows, and build an explanation from their own evidence.

Keywords: Astronomy education (2165) — Astronomy software (1855) — Computational methods (1965) — Open source software (1866) — Physics education — Interactive learning environments

1. INTRODUCTION

Astronomy is unusually rich in phenomena that students can see but cannot directly manipulate. They cannot move a planet closer to its star, turn off a galaxy’s dark matter, replay a close stellar encounter, or wait through a main-sequence lifetime. Even familiar ideas such as an elliptical orbit or a transit light curve require students to coordinate several representations at once: a moving physical system, a graph, a mathematical relationship, and the viewpoint of an observer.

Interactive simulations can make such systems available for inquiry when the simulation is paired with purposeful prompts, measurements, and opportunities to explain the result (Finkelstein et al. 2005; Wieman et al. 2008; Rutten et al. 2012). Active engagement is not created by animation alone; students need a question to an-

swer and evidence with which to answer it (Hake 1998; Freeman et al. 2014). GRAVITAS was built around that distinction. It can be opened as a free-form sandbox, but it also contains a complete instructional layer that asks students to predict an outcome before running a system, measure the result with task-specific instruments, compare it with their prediction, and explain what changed.

The intended audience is broad: instructors teaching introductory astronomy or physics, students exploring independently, and presenters who want a reproducible live demonstration. A visitor needs no account, installation, or specialized hardware. The simulation runs in a modern web browser at <https://gravitas-sim.online>; the source is openly available under the MIT License. A complete state can be encoded in a URL, allowing an instructor to distribute the same initial conditions to an entire class and allowing a student to return a modified system as part of an answer.

This paper presents GRAVITAS as a teaching environment rather than as a software architecture. Section 2

introduces the student experience. Sections 3 and 4 describe the guided investigations and measurement tools. Section 5 follows three representative learning experiences, while Section 6 outlines practical classroom use. Section 7 states what the models can and cannot support. The paper describes the release candidate represented by source commit `a4e6ab3`; counts and screenshots should be refreshed from the tagged 1.0.0 release before submission.

2. FROM FIRST CLICK TO EXPERIMENT

GRAVITAS opens directly into an interactive system. The central canvas shows the evolving scene, while controls expose bodies, scenario settings, time, viewpoint, and measurement tools. A learner can start from the Solar System, TRAPPIST-1, a binary star, a spacecraft transfer, a galactic rotation curve, or a black-hole encounter, then change the masses, positions, velocities, and physical options that define the experiment. The current gallery contains 59 systems spanning orbital motion, exoplanets, stars, galaxies, compact objects, and playful counterfactual cases. Concept tags let an instructor find a system that matches the topic of a particular class meeting.

The interface is designed to move naturally between qualitative observation and quantitative evidence. Trails reveal orbital shape and motion. Velocity and acceleration arrows expose the changing geometry of force and motion. A selection inspector reports mass, radius, speed, orbital elements, and energy for an individual body. A ruler, protractor, stopwatch, plots, and event-aware measurements let students attach numbers to what they see. The elapsed simulation time and spatial scale remain visible, and exported screenshots carry the scenario name, clock, and scale so that an image remains interpretable after it leaves the application.

A scenario is not a locked animation. Students can pause, step, rewind by restoring a captured state, or change the rate at which simulated time passes. They can follow a body with the camera or re-express the entire recorded scene in that body’s reference frame. The latter distinction matters: changing to Earth’s frame redraws the prior trails and causes Mars to trace the familiar retrograde loop, while the underlying heliocentric dynamics remain unchanged. The student therefore sees reference frame as a choice of description rather than a change in physical law.

The same emphasis on reproducibility appears throughout the application. Shareable links preserve the initial configuration. Seeded scenarios reproduce the same controlled trial. An A/B experiment bench captures a state, records a baseline, restores the identi-

cal state, and then records a second run after one variable is changed. The resulting comparison names the changed parameter and can be exported with provenance. These features support a simple but important classroom habit: change one thing, preserve everything else, and decide whether the evidence supports the claimed cause.

3. GUIDED INVESTIGATIONS

The 22 guided investigations contain 631 student-facing steps. They range from short, 15–25 minute spaceflight activities to 80–100 minute stellar-evolution sequences. All are aimed primarily at introductory astronomy, with one gravitational-wave investigation explicitly requiring no prior physics.

Each investigation follows a recurring learning cycle:

1. **Predict.** The learner commits to an expected outcome before the simulation reveals it. Predictions are recorded but are not graded for correctness.
2. **Test.** A prepared scene loads with the controls and instruments needed for the question.
3. **Measure.** The learner records values from the running model rather than copying numbers from explanatory text.
4. **Revise.** The learner compares the result with the original prediction and, where appropriate, changes a variable or repeats a controlled run.
5. **Explain.** A short response, calculation, graph, or comparison asks the learner to connect the evidence to the physical idea.

This structure is intended to limit passive watching. The prediction gives the result something to challenge; the measurement makes the conclusion auditable; and the final explanation requires the student to say why the result occurred. Progress is stored locally, and the finished evidence notebook can be exported as a PDF for an existing learning-management system. No student account or external data service is required.

Table 1 summarizes the breadth of the catalogue. The investigations are not merely tours of features: each is organized around a claim that students can settle with a measurement. For example, the resonance activity first demonstrates that a near-integer period ratio is not sufficient evidence for resonance, then asks students to distinguish libration from circulation of the resonant angle. The binary-planet activity maps the boundary between surviving and disrupted orbits. The gravitational-wave sequence asks students to distinguish what is physically calculated, what is an illustrative visualization, and what an observer measures.

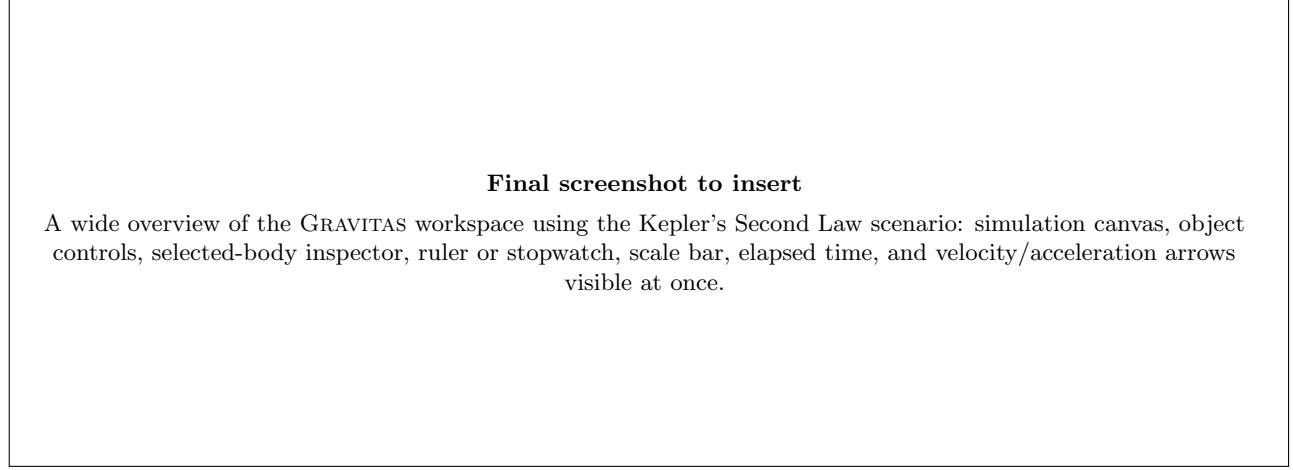


Figure 1. The GRAVITAS workspace is both a visual sandbox and a measurement environment. The final image should show a scientifically legible scenario rather than a decorative collision: the orbit, clock, scale, vectors, and at least one measurement tool should all be visible.

Table 1. Guided investigation catalogue

Theme	Investigations	Representative student work
Orbits and gravity	Kepler’s Laws; Why Mars Goes Backwards; Bound, Unbound and Escape; Weighing the Stars; Tides	Locate the focus of an ellipse; compare equal swept areas; transform reference frames; classify trajectories by energy; infer stellar mass from a binary orbit; measure differential gravity.
Exoplanets and observing	Finding Planets by Their Shadows; Finding Planets by Their Tug; The Goldilocks Question; Can You Detect This Planet?; Design the Schedule; Planets in Binary Stars	Recover radius and period from a transit; infer mass from radial velocity; combine mass and radius; explore inclination and dilution; design observations; map stable regions around one or two stars.
Orbital dynamics and spaceflight	The Butterfly Effect in Space; When Orbits Lock; Where Does a Gravity Assist Get Its Speed?; Getting There From Here; Where Can It Get To?	Measure divergence of nearby starts; test resonant angles; compare a flyby in two frames; calculate and fly a Hohmann transfer; use the Jacobi constant and zero-velocity regions.
Compact objects and gravitational waves	Black Holes by the Numbers; What Is a Gravitational Wave?; Listening to Spacetime	Test how black-hole properties scale with mass; connect orbital motion to a travelling strain; measure a chirp and compare the model with GW150914 (Abbott et al. 2016).
Stars, galaxies, and dark matter	A Universe of Stars; Lives of Stars; The Missing Mass	Separate mass, radius, temperature, and luminosity on the H–R diagram; follow published stellar tracks (Dotter 2016; Choi et al. 2016); fit a rotation curve and compare luminous and dynamical mass.

Note. Durations and prerequisites are supplied in the application. The catalogue can be filtered by concept, and individual investigations may be assigned through reproducible links.

4. TOOLS THAT TURN MOTION INTO EVIDENCE

The tools in GRAVITAS are selected for questions commonly asked in introductory astronomy. They are available in the sandbox and are also opened automatically when a guided step requires them.

Three observing panels are especially useful for connecting dynamics to astronomical inference. A transit

light curve, a radial-velocity curve, and an astrometric track are generated from the same evolving system and the same observer geometry. Moving the observer therefore changes all three consistently: a planet can cease to transit, the radial-velocity amplitude changes with inclination, and the astrometric line opens into an ellipse. Students encounter selection effects and degeneracies as

Table 2. Student tools and the reasoning they support

Tool	What the student records	Example question
Ruler, protractor, and scale bar	Distances in astronomical units or kilometres; angles between locations or vectors	Where is the focus? How far does a planet move in a fixed interval? What angle separates a Trojan asteroid from Jupiter?
Simulation-time stopwatch	Elapsed time, including automatic latching at successive periapsis passages	How can a period be measured without relying on reaction time?
Object inspector	Mass, radius, speed, orbital elements, and kinetic, potential, and total energy	Is the trajectory bound? Which body is the orbit actually referenced to?
Velocity, acceleration, and source arrows	The selected body's velocity, total acceleration, and contribution from each gravitating source	Why is an orbit curved? Why are velocity and force not generally parallel?
Reference-frame controls	The same positions and recorded trails relative to the barycenter or a selected body	Why does Mars reverse direction in an Earth-centered view?
A/B experiment bench	Aligned baseline and comparison time series, changed parameter, CSV, and provenance metadata	Does changing the starting distance, rather than the random seed or timestep, cause the different outcome?
Observing panels	Transit flux, radial velocity, astrometric displacement, viewing geometry, and sampled observations	Which properties of a planet can each method recover, and how does inclination change the evidence?
Specialized plots	Swept area, rotation curve, resonant angle, stellar tracks, gravitational-wave strain, and conservation diagnostics	What observation distinguishes a resonance from a near-integer ratio? Why does a galaxy require more gravitating mass than its light implies?
Exports	Screenshots, clips, report PDF, state link, and unit-labelled CSV	Can another student or instructor reproduce the setup and inspect the evidence?

Final screenshot to insert

A vertically arranged composite from one investigation showing (1) a prediction prompt, (2) an attached live measurement, and (3) the evidence/report summary. Use one continuous example, preferably Kepler's Laws or Finding Planets by Their Shadows.

Figure 2. A guided investigation turns interaction into a record of reasoning. Students predict before the run, attach measurements from the live scene, and finish with an exportable evidence-based explanation.

outcomes of a controlled experiment, rather than only as warnings appended to a formula.

The application also makes numerical reliability visible when it is relevant. Students may choose among

three integrators and inspect changes in total energy and angular momentum. Dedicated investigations distinguish a physical sensitivity to initial conditions from ordinary phase drift or numerical error. The purpose is not to teach numerical analysis in every activity; it is to let students ask whether a surprising result survives an appropriate reliability check.

5. THREE PATHS THROUGH THE SIMULATION

5.1. *Discovering Kepler's Laws*

In the Kepler investigation, students begin with the geometry of an ellipse. They locate the star relative to the two foci, alter eccentricity, and compare circular and eccentric orbits. The simulation then displays equal-area sectors over equal time intervals while the student independently measures the fastest and slowest portions of the orbit. Finally, periods and semimajor axes from multiple systems are plotted to recover a slope of $3/2$ in log space.

This sequence connects picture, measurement, and equation. The second law is not reduced to the statement that a planet moves faster when it is closer. The learner sees the swept area, times the motion, and relates the result to angular momentum. The third law is not supplied as an exponent to verify; it is the re-

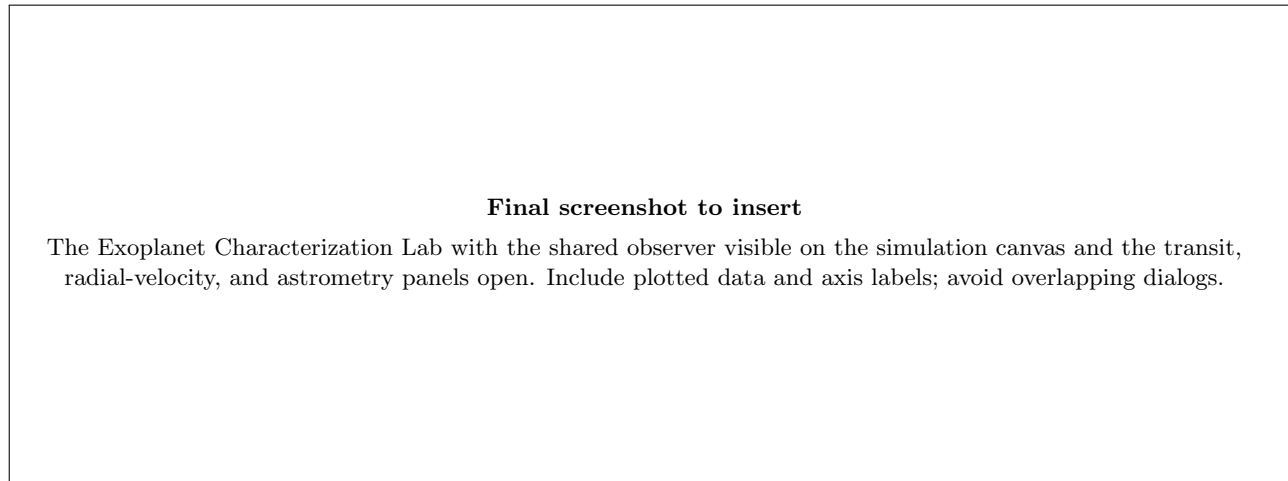


Figure 3. One simulated planetary system viewed through three observational methods. Because the panels share the same observer, changing the viewing geometry immediately exposes which inferences are robust and which depend on inclination or alignment.

lationship that emerges from the student’s graph. The same content can be used as a five-minute prediction and demonstration, a roughly twenty-minute guided activity, or a full laboratory investigation.

5.2. *Becoming an Exoplanet Observer*

The exoplanet sequence begins with the shadow of HD 209458 b, a landmark transiting system (Charbonneau et al. 2000). Students measure a transit depth, infer a radius, examine limb darkening, determine a period, and explore how blended light from a companion star biases the result. The radial-velocity investigation then uses the star’s reflex motion to estimate the planet’s mass. Combining the two measurements produces a bulk density and a more physically meaningful description of the planet.

Later investigations shift the question from parameter recovery to experimental design. Students receive a finite number of observing nights, choose a cadence, and compare schedules under the same signal and noise realization. They see how aliasing, weather losses, baseline, and sampling affect whether a planet is detected. The Goldilocks investigation then changes orbital distance, stellar luminosity, and eccentricity to examine incident flux and the limits of the phrase habitable zone, using established habitable-zone relations as the scientific context (Kopparapu et al. 2013).

Together, these activities reproduce the logic of observation: no single panel reveals everything, geometry matters, and the observing strategy can determine which conclusions are possible.

5.3. *When the Visible Matter Is Not Enough*

The Missing Mass investigation asks students to weigh systems twice. First they sum the matter that is visible; then they infer gravitating mass from orbital motion. These estimates agree in a Solar System-like case but diverge for a galaxy and a galaxy cluster. Students arrange visible mass, inspect the rotation curve it produces, and attempt to reproduce a measured flat curve. Adding an extended halo changes the force law used by the orbiting tracers and allows the curve to remain flat at large radius, connecting the experiment with modern rotation-curve data sets such as SPARC (Lelli et al. 2016).

This activity illustrates a broader goal of GRAVITAS: students should experience why an inference was made. The conclusion that galaxies contain dark matter becomes the resolution of a failed mass model rather than a fact presented without its observational pressure.

6. DESIGNED FOR REAL CLASSROOMS

An educational tool is only useful if it fits the time, equipment, and workflow available to an instructor. GRAVITAS supports several levels of use:

- **Live demonstration.** Lecture mode enlarges text, selects a high-contrast daylight theme, supplies a spotlight pointer, and lets an instructor step through a prepared sequence with arrow keys.
- **Embedded figure.** A share link with embed mode produces a chrome-free interactive view suitable for a course page or learning-management system.
- **Short activity.** Curated routes extract a coherent prediction–experiment–explanation cycle from

a longer investigation for use during a single class meeting.

- **Laboratory assignment.** A full investigation saves progress and produces a student report that can be submitted through the course’s existing system.
- **Open exploration or project.** The sandbox, A/B bench, share links, and data export support questions designed by the learner or instructor.

For each guided investigation, the instructor collection includes learning objectives, expected observations, common wrong turns, and an answer key generated from the same lesson source used by students. An adopter’s guide and curriculum map help place activities into an existing course. Public teaching pages explain the workflow and provide demonstration links before an instructor requests access to protected answer materials.

Access considerations are treated as instructional infrastructure rather than optional polish. The interface and all investigations are available in English and Spanish. Keyboard navigation, screen-reader labels, reduced-motion behavior, multiple themes, and automated accessibility checks support a wider range of learners. The application can operate offline after its assets have been cached, including on lower-powered devices with a reduced visual-quality tier. Because all computation and progress storage occur locally, students are not required to create an account or send their work to a third-party service.

These features also make assignments more resilient. A link can reproduce the starting state across a large class. A screenshot documents its own time and scale. A CSV names the units in its columns. A report gathers the student’s predictions, measurements, graphs, and explanations. The goal is to reduce the amount of custom technical scaffolding an instructor must build before the physics can begin.

7. SCIENTIFIC SCOPE, TRANSPARENCY, AND VALIDATION

GRAVITAS is an educational model, not a professional research code. Its core dynamics are Newtonian and two-dimensional. Point-mass gravity is integrated in a plane; close encounters use softening; and some scenarios allow selected bodies to feel gravity without exerting it so that large tracer populations remain responsive. Collisions are represented as perfectly inelastic mergers. Relativistic quantities such as Schwarzschild radius may be evaluated analytically, but general-relativistic orbital dynamics are not solved. Gravitational-wave in-

spiral uses an educational prescription, and visual elements such as accretion disks, jets, and the optional rubber-sheet spacetime view are illustrative. Body sizes are compressed for visibility and must not be measured from their drawn diameters.

These boundaries are stated prominently on the public model page. Features are labelled as simulated, analytic, approximated, illustrative, or not modeled. That vocabulary is also pedagogically useful: students can ask whether a particular conclusion follows from the evolving equations, a closed-form relationship, or a visual analogy. Instructors can decide whether the model is appropriate for the learning objective before assigning it.

A public validation report tests orbital periods, Kepler relationships, conservation laws, convergence behavior, escape speed, exoplanet observables, habitable-zone quantities, rotation curves, black-hole scaling relations, and stored parameters for real systems against stated expectations or published sources. Scenario-stability checks audit the shipped catalogue, while browser tests exercise the student workflows. These checks cannot establish that the software is error-free, nor can they substitute for classroom evaluation. They do make the scientific claims, tolerances, known approximations, and failure conditions inspectable.

Version 1.0.0 should therefore be evaluated along two separate dimensions. Scientific verification asks whether the model produces the quantity it claims to produce within an appropriate tolerance. Educational evaluation asks how students and instructors use the environment and whether the intended reasoning appears in their work. The first is included in the release process. Formal evidence for the second is future work. The teaching site provides a local feedback template so adopters can record course context, time on task, friction points, and observed student reasoning without transmitting student data.

8. AVAILABILITY AND REUSE

The simulation is available at <https://gravitas-sim.online>. Source code, documentation, model limitations, validation materials, and issue tracking are available at <https://github.com/gravitas-sim/gravitas-sim.github.io>. The project is released under the MIT License.

The archival record for version 1.0.0 will be deposited with Zenodo before the final manuscript is submitted:

DOI: ZENODO-DOI-PLACEHOLDER

Zenodo record: ZENODO-RECORD-URL-PLACEHOLDER

The placeholders above are intentionally explicit so they can be found and replaced after the GitHub release

Final screenshot to insert

A clean four-panel montage showing the scenario gallery plus three contrasting learning environments: the Missing Mass rotation-curve view, Listening to Spacetime or GW150914, and the stellar H–R diagram. Use labels (a)–(d) and retain readable axes.

Figure 4. GRAVITAS extends beyond a conventional orbit sandbox. The final montage should communicate both the searchable breadth of the scenario gallery and representative activities in galactic dynamics, gravitational waves, and stellar astrophysics.

has been archived. The tagged release, Zenodo record, manuscript version, scenario and investigation counts, and final screenshots should all describe the same 1.0.0 state.

Instructors may use the built-in investigations as written, assign selected steps, or create their own activities from shared scenarios and exported data. The repository welcomes reports from classroom use, translations, new scenarios, and proposed investigations. The most valuable feedback is concrete: what learners predicted, what they measured, where an interface choice interrupted the reasoning, and what explanation emerged after the experiment.

9. CONCLUSION

GRAVITAS gives students a laboratory for systems that ordinary laboratories cannot hold. A planet can be moved, a viewpoint changed, a dark halo removed, a binary replayed from the same initial state, or an observing schedule redesigned within minutes. More im-

portantly, the environment asks students to make those changes for a reason: to test a prediction and construct an explanation from measured evidence.

For an instructor, the entry cost is deliberately low: open a link, choose a scenario or investigation, and use it as a projected demonstration, an embedded interactive figure, a short activity, or a complete lab. For a student, the invitation is equally direct: build a system, run it, measure it, and find out whether the universe in the model behaves as expected. GRAVITAS is available now at <https://gravitas-sim.online>, and the versioned 1.0.0 archive will be citable at the DOI above when the release is minted.

- 1 The author welcomes feedback from instructors and stu-
- 2 dents who use GRAVITAS in a course. Development and
- 3 maintenance information will be added here before sub-
- 4 mission if institutional or funding acknowledgment is re-
- 5 quired.

Software: GRAVITAS (Ziegler 2026)

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